

The Sepsis Six: Does Public Data Show Whether It Worked?

Bootstrap CUSUM Analysis — ONS Sepsis Mortality England and Wales 2001–2023 — Executive Summary

stepchangeanalysis.com | May 2026 | Syd Stewart, Chartered Chemical Engineer | Data: [sepsis-both-series.csv](#)

Bootstrap CUSUM asks whether a time series structurally changed at any point and when. It resamples the data thousands of times to test whether an apparent change point is real or random variation. N=23 annual observations (2001–2023). England and Wales population grew 15% 2001–2023; population ageing increased sepsis-susceptible people independently of clinical outcomes.

Five Key Findings

1

The Sepsis Six — six actions, one hour, 96% of NHS hospitals

Developed in 2005 by Dr Ron Daniels (Good Hope Hospital) from the Surviving Sepsis Campaign guidelines. Six bedside actions within one hour of suspected sepsis: (1) High-flow oxygen — tissue oxygenation; (2) Blood cultures — before antibiotics; (3) IV antibiotics — every hour of delay increases mortality; (4) IV fluid resuscitation — 500ml crystalloid; (5) Serum lactate — tissue hypoperfusion marker; (6) Urine output — renal perfusion monitor. Now endorsed by NICE, used in 96% of NHS trusts and 37 countries. UK Sepsis Trust 2011 study: 50–54% mortality reduction with full compliance vs non-compliance.

Developed: 2005 | NHS CQUIN adoption: 2013 | NICE NG51: 2016 | Mandatory screening: 2017

2

Mentioned anywhere series: one stage, no change point across 22 years

Deaths where sepsis was mentioned anywhere on the death certificate — the most complete clinical measure, capturing the 91.4% of sepsis deaths not coded as primary cause. N=23, Mean=23,556, SD=2,566. Bootstrap CUSUM: Stages=1, no change point at 90% confidence. X-mR: all values within natural process limits (UNPL=27,069, LNPL=20,043). Run chart: shift signal 6+ above median ~2004–2009 (Surviving Sepsis Campaign era). No shift signal during NHS Sepsis Six rollout 2012–2015 or Think Sepsis 2017.

Stages=1 | No change point | Mean=23,556 | UNPL=27,069 | LNPL=20,043

3

Underlying cause series: one stage, no change point — dominated by coding changes

Deaths where A40/A41 was the primary underlying cause — only 8.6% of true sepsis deaths. N=23, Mean=2,488. Bootstrap CUSUM: Stages=1, no change point. Run chart: 6 consecutive points below median 2007–2013 (under-coding era), then sharp sustained rise from 2014 (Think Sepsis coding improvement). The rising headline number from 2014 reflects better coding recognition, not more deaths.

Stages=1 | No change point | Mean=2,488 | 6-point shift below median 2007–2013

4

The ratio: structural change point in 2013 at 95.5% confidence — the smoking gun

Bootstrap CUSUM on the ratio (mentioned anywhere / underlying cause): Stage 1 (2001–2013): mean ratio 10.75. Stage 2 (2013–2023): mean ratio 8.63. Change: -19.7% at 95.5% confidence. The measurement system changed structurally in 2013 — precisely when the NHS CQUIN financial incentive for sepsis screening was introduced. Better coding. Not better or worse clinical outcomes. The Bootstrap CUSUM finds no change point in either clinical series but finds a significant change point in the ratio — the most precise possible statement of why public data cannot evaluate the Sepsis Six.

Ratio Stage 1 mean: 10.75 | Stage 2 mean: 8.63 | Change: -19.7% | Confidence: 95.5%

5

What this means for clinical QI — the PDSA Study step was never completed

PDSA (Plan-Do-Study-Act) requires pre-specifying the outcome before the intervention: "We expect this to produce a detectable Bootstrap CUSUM change point in [measure] within [X] years." The Sepsis Six was adopted nationally without this pre-specification, and the outcome measure (underlying cause mortality) changed during the rollout due to coding campaigns. Three measures would actually answer the question: (1) HES in-hospital mortality rate — requires NHS Digital DARS request; (2) NHS England sepsis audit data (time-to-antibiotics, bundle compliance) — available 2018+, in 5–7 years will be sufficient for Bootstrap CUSUM; (3) Age-standardised mortality rate — corrects for population ageing, not currently available for this series.

PDSA Study step incomplete | HES data needed | Audit data: sufficient for CUSUM by ~2030

Three Bootstrap CUSUM Analyses — Summary

Series	Stages	Change point	Confidence	What it means
Mentioned anywhere deaths	1	None	—	No structural change in total sepsis-associated mortality across 22 years
Underlying cause deaths (A40/A41)	1	None	—	No change point — rising trend 2014+ is coding improvement not clinical change
Ratio (mentioned / underlying)	2	2013	95.5%	Coding improvement structural change point coincides with CQUIN introduction. -19.7% in ratio.

Recognition Campaigns — Timeline

Year	Campaign / Intervention	Effect on coding
2002+	Surviving Sepsis Campaign — international guidelines	Raised ICU/ED awareness. Limited ward-level coding impact.
2013	NHS England CQUIN — financial incentive for sepsis screening	Direct financial incentive to document sepsis. Bootstrap CUSUM ratio change point at 95.5%.
2016	NICE NG51 — first NICE sepsis guideline	Extended focus to all clinical settings. Red Flag Sepsis concept.
2017	Think Sepsis — mandatory A&E; and acute screening	Largest coding-level intervention. All acute admissions formally screened and documented.

What Would Actually Show Whether the Sepsis Six Worked

Measure	Available	Why it would work
HES in-hospital mortality rate for sepsis admissions	NHS Digital DARS request	Captures clinical outcomes not coding. In-hospital mortality as % of sepsis-coded admissions. Available back to 2003. The correct primary measure.
NHS England sepsis audit data (time-to-antibiotics, bundle compliance)	Public 2018+ N=6 currently	Lead measures predicting future outcomes. In ~5 years will have N=11+, sufficient for Bootstrap CUSUM. The prospective test.
Age-standardised mortality rate for A40/A41	Needs ONS data request	Removes population ageing effect. Not available in current dataset. Would partially address coding change problem.



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Key References
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Data file: sepsis-both-series.csv — stepchangeanalysis.com